



TG_3SIXO

Team - Route()Proof

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Introduction

What is TG_3SixO

It is a smart transport delay-forecasting system.

It fuses official data, community reports, and live bus feeds to forecast upcoming delays.

Results are shared as simple, open data for apps, maps, and passengers.

Why TG_3SixO

Reactive apps show delays after they happen.

Journey Radar predicts them before they do.

Who It's For

- **Commuters:** Plan smarter and travel stress-free
- **Dispatchers / Operators:** Get early disruption alerts and send warnings to others on route
- **City systems & apps:** Integrate predictive data to improve routing and planning
- **Drivers:** Stay updated with real-time road and route conditions



Challenge	Journey Radar Solution
Uncertainty “Is my bus late?”	Provides forecasted delay windows with confidence levels so you know before it happens.
Slow updates	Uses community-verified alerts to fill the gap before official operator updates appear.
Low engagement	Introduces a points and rewards system for accurate reports. Trust scores unlock multipliers and partner perks (e.g. Żabka).
Scattered data	Combines GTFS feeds, user input, live reports, and dispatcher data into one unified, reliable view.

 **Example:** At 07:30, TG_3SixO predicts Bus 152 will be 12 minutes late at 09:30 with 96 % confidence.

 Impact

Faster communication • Less passenger stress • Smarter dispatching
TG_3SixO transforms delay management into delay prevention.



🧩 User-Reported Disruptions & Reward System

Our Goal: Turn passengers into real-time sensors for city transport.

- Instant Reporting — Users submit delays or accidents with title, photo, and severity; reports appear live on the map.
- Smart Impact — Every verified report feeds our TransitDNA engine, dynamically adjusting route predictions in real time.
- Verification & Rewards — Dispatchers validate reports; early and accurate users earn reward points redeemable via partners (e.g. Żabka or discounted tickets).
- Emergency Escalation — Critical incidents trigger dispatcher calls ( /  / ) with shared GPS and contact data.
- Tech Backbone — Python (FastAPI) + C++ core ensure live syncing, verified storage, and instant ETA recalculation.

Predictive Delay Intelligence

Data Fusion Engine — Combines official GTFS feeds, historical delay data, and real-time user reports into a single unified model.

Hybrid Forecast Core

- C++ backend for live simulation
- Python ML pipeline for pattern recognition and delay prediction

Real-Time Adaptation — Continuously learns from new incidents and adjusts forecasts as city conditions change (events, accidents, etc.)

Actionable Insights — Predicts delay windows (e.g., “+12 min expected at 09:30”) so dispatchers and commuters can act before disruptions occur

⚡ Real-Time Delay Intelligence

Live Data Streams — Integrates continuous GTFS vehicle feeds, user reports, and simulated network data for instant updates.

Smart Map Visualization — Interactive map highlights affected routes, delay severity, and estimated recovery time in real time.

Dynamic ETA Recalculation — C++ backend recalculates routes and travel times instantly as new incidents are detected.

Personalized Context — Adjusts delay info based on the commuter's current location and saved journeys.

Seamless Server Push — Uses Server-Sent Events (SSE) to proactively push alerts the moment a new delay or reroute occurs.

Dispatcher Systems

Integrated API Interface — Lightweight REST endpoints let dispatchers fetch, verify, or post disruptions in real time.

Two-Way Communication — Dispatchers can receive passenger reports, confirm incidents, and push verified alerts instantly into the live network.

Emergency Coordination — Supports dispatcher calls with service tags ( Police /  Ambulance /  Firefighters) and live GPS sharing for rapid response.

Smart Data Integration — Ingests real-time sensor data from vehicles (speed, stop duration, GPS) to auto-detect delays and validate user reports.

Predictive Modelling — Combines historical and user-submitted data to train forecasting models, improving prediction accuracy over time.



Maps & Navigation

Interactive Map Interface — Shows live disruptions, incidents, and reroutes using color-coded severity markers.

Optimal Route Recalculation — Dijkstra-based C++ routing engine dynamically avoids affected nodes and recomputes ETA.

Multi-Source Fusion — Merges GTFS data, user submissions, and system forecasts into one consistent visual.

Persona-Based Navigation — Personalized for saved journeys (e.g., “Sarah’s daily commute”), showing tailored delay predictions.

Real-Time Visualization — Map and UI update instantly via Server-Sent Events, giving passengers proactive alerts.

Backend Tool Stack



Server Core (C++)

- Hosts backend endpoints & real-time SSE events
- Handles route requests, calendar imports, and user reports
- Ensures uptime and concurrency using threads + mutex locks

Predictive System (TransitDNA)

- Brain of TG_3SixO — learns from historical and live disruptions
- Uses rolling averages + real-time data to forecast ETA delays
- Suggests alternate routes when risk thresholds are exceeded

Transit Engine (GTFS Integrator)

- Connects to official Kraków GTFS feeds (bus & tram)
- Normalizes live trip & vehicle updates and merges them with user/community reports
- Powers the real-time travel intelligence pipeline

Delay Forecasting (ML-Ready)

- Built for integration with Chronos time-series models
- Generates predictive delay windows (e.g., “96% chance of +12 min delay”)
- Continuously retrain models as new reports and GTFS data arrive



Frontend Tool Stack — Live Map & User Experience

Real-Time Dispatch Dashboard

- Displays live vehicle data and predicted disruptions
- Syncs with backend so dispatchers act before delays cascade

Commuter App UI

- Built on Leaflet.js SPA — clean map, modern visuals
- Shows predicted delays, probabilities, route alternatives, and rewards

Community & Reward Layer

- Users report incidents (accidents, jams, blockages) via one-tap form
- Verified reports earn points redeemable for partner rewards (e.g., Žabka, discounted tickets)

Calendar & Persona Smart Tools

- Calendar conflicts warn users if ETAs clash with meetings
- Persona system (e.g., “Sarah”) learns habits and predicts personal delay risks